

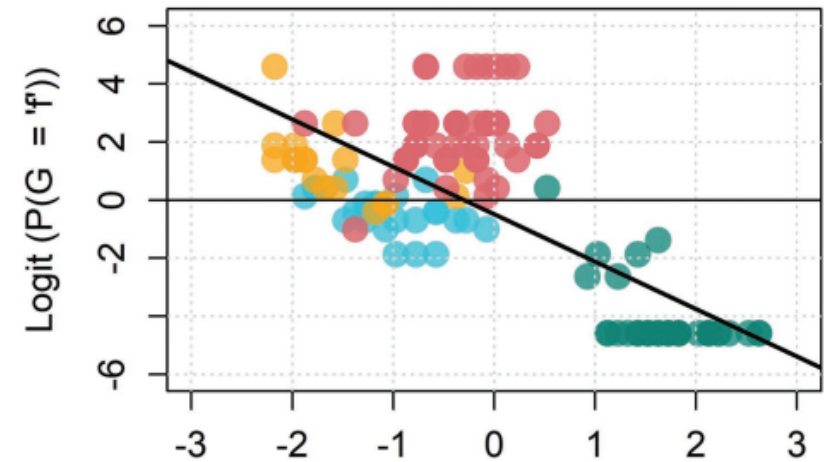
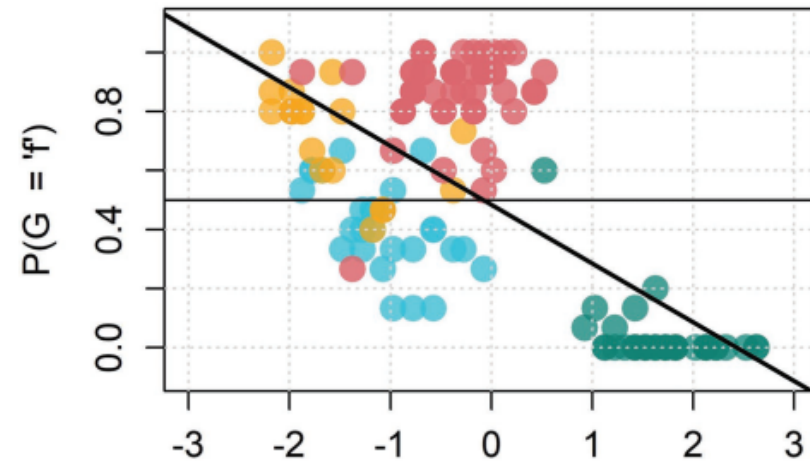
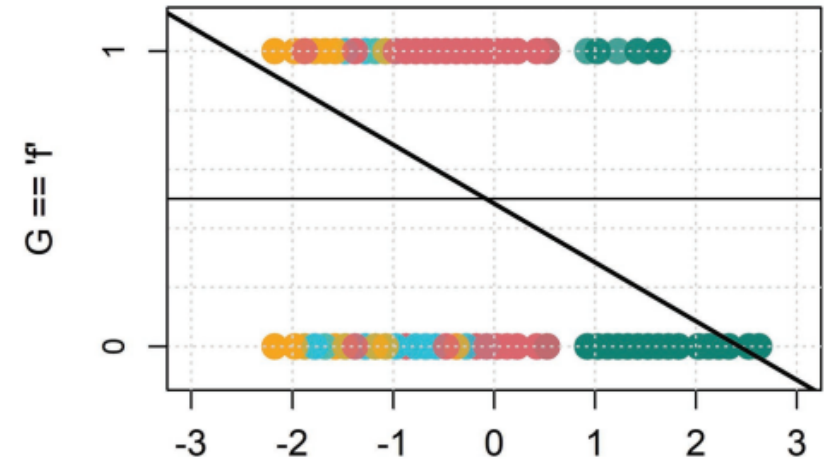
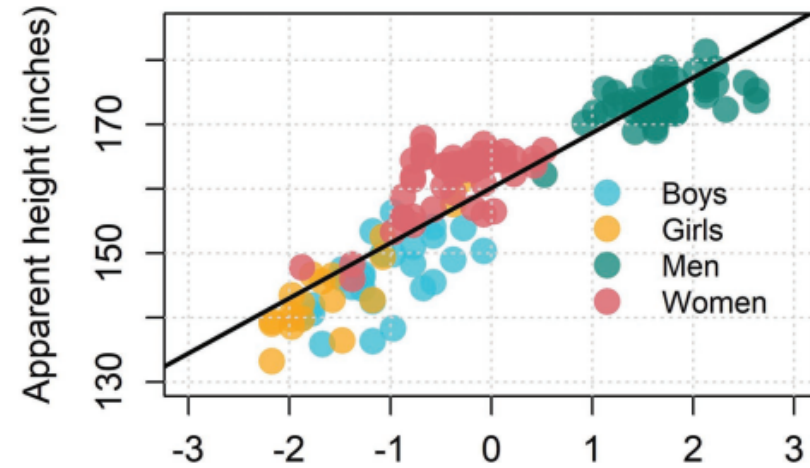
# Capítulo 10

# Chapter Precap

- Modelos lineales para la predicción de variables dicotómicas que pueden tomar uno de dos valores discretos posibles.
- Introducimos datos dicotómicos y las distribuciones Bernoulli y binomiales.
- Se discuten las funciones de enlace y el modelo lineal generalizado.
- Se introducen la regresión logística, los logits y la función logit inversa.
- Ajustamos un modelo de regresión logística multinivel con un único predictor cuantitativo.
- Usamos modelos para comprender predicciones y mapas territoriales.
- Introducimos el uso de la regresión logística para ajustar un modelo de teoría de detección de señales, estimando el sesgo de respuesta y la sensibilidad utilizando un modelo de regresión logística multinivel.

# Linear Models and Dichotomous Data

$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + VTL \cdot \text{vtl}_{[i]}$$



Centered VTL (cm)

# Bernoulli and Binomial Data

```
# a single trial, probability of 0.5
bmmb::rbernoulli (1,.5)
## [1] 0

# ten single trials, probability of 0.5
bmmb::rbernoulli (10,.5)
## [1] 0 0 0 0 0 0 0 1 0 1 0
```

```
# a single batch of 10 trials, probability of 0.5
rbinom (1,10,.5)
## [1] 6

# ten individual trials, probability of 0.5
bmmb::rbernoulli (10,.5)
## [1] 1 1 0 0 0 0 0 0 0 1
```

# Bernoulli Expected Value

$$\mathbb{E}(y) = \sum_{i=1}^2 y_{[i]} P(y_{[i]})$$

$$\mathbb{E}(y) = (1 \cdot p) + (0 \cdot (1 - p))$$

$$\mathbb{E}(y) = p$$

```
mean (rbernoulli (10, .5)) # the mean of 10 observations
## [1] 0.3
mean (rbernoulli (100, .5)) # the mean of 100 observations
## [1] 0.47
mean (rbernoulli (1000, .5)) # the mean of 1000 observations
## [1] 0.488
mean (rbernoulli (100000, .5)) # the mean of 100000 observations
## [1] 0.4999
```

# The Generalized Linear Model

- Los modelos de regresión se componen de:

- Un componente aleatorio

$$\text{height}_{[i]} \sim \mathcal{N}(\mu_{[i]}, \sigma)$$

- Un componente sistemático.

$$\mu_{[i]} = \text{Intercept} + VTL \cdot \text{vtl}_{[i]}$$

- Una función de enlace...

# The Generalized Linear Model

- 1 Predict variation in expected values ( $\theta$ ) along straight lines (or related shapes), for example  $\theta = a + b \cdot x$ . This is the systematic component and  $\theta$  is linearly related to the predictor  $x$ .
- 2 Transform  $\theta$  using a link function. For example, given the link function  $f$  we can transform our expected value  $\theta$  to  $p$  like so:  $p = f(\theta)$ . The transformed parameter ( $p$ ) may no longer be linearly related to the dependent variable  $x$  or the linear predictor  $\theta$ .
- 3 Use the *transformed* parameter ( $p$ ) in the data generating distribution. For example  $y \sim \text{Bernoulli}(p)$ . This is the random component.

$$\text{Female}_{[i]} \sim \text{Bernoulli}(p_{[i]})$$

$$p_{[i]} = f(\theta_{[i]})$$

$$\theta_{[i]} = \text{Intercept} + VTL \cdot \text{vtl}_{[i]}$$



# Logistic Regression

Regresión logística:

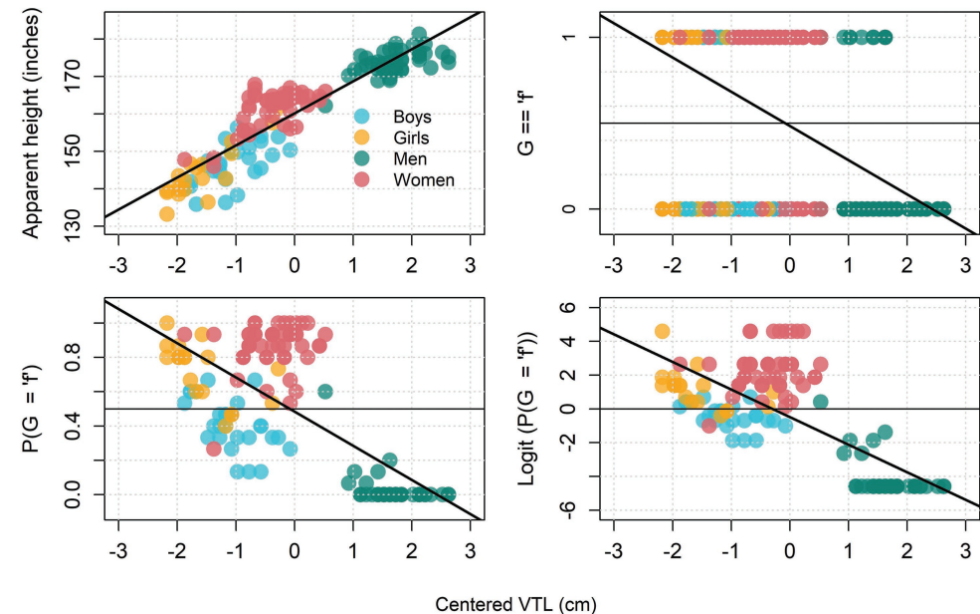
- Predice un valor esperado en logits (z) dada alguna combinación de los predictores.
- Convierte este valor esperado de logits a probabilidades (p).
- Los datos provienen de una distribución de Bernoulli con un parámetro igual a p.

$$y_{[i]} \sim \text{Bernoulli}(p_{[i]})$$

$$p_{[i]} = \text{logit}^{-1}(z_{[i]})$$

Función de  
enlace logit  
inverso

$$z_{[i]} = \text{Intercept} + \beta \cdot x_{[i]}$$





# Odds

Las probabilidades de éxito: la relación entre el número de éxitos y el número de fracasos.


$$\text{odds}_{\text{success}} = \frac{N_{\text{success}}}{N_{\text{failures}}}$$

Para convertir probabilidades en cuotas.


$$\text{odds} = p / (1 - p)$$

Para convertir cuotas en probabilidades.


$$P(\text{success}) = \frac{N_{\text{success}}}{N_{\text{failures}} + N_{\text{success}}} = \frac{N_{\text{success}}}{N_{\text{total}}}$$

# Logits

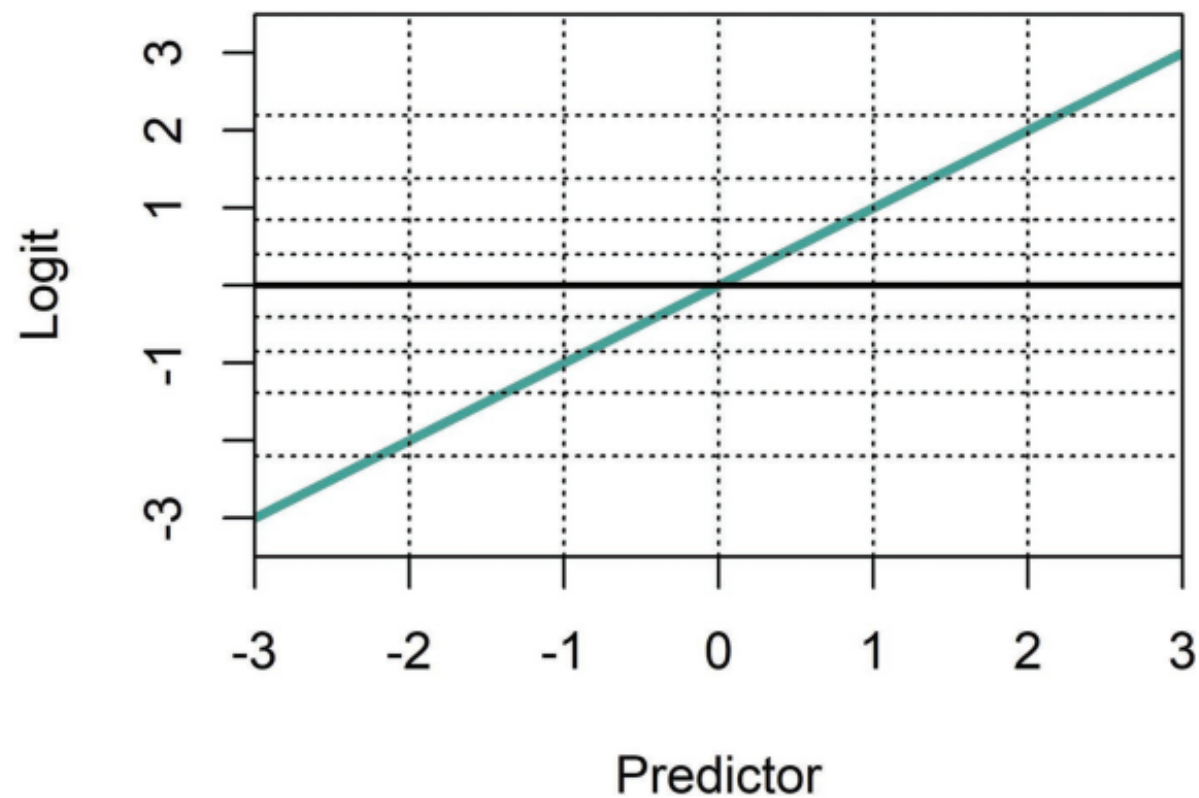
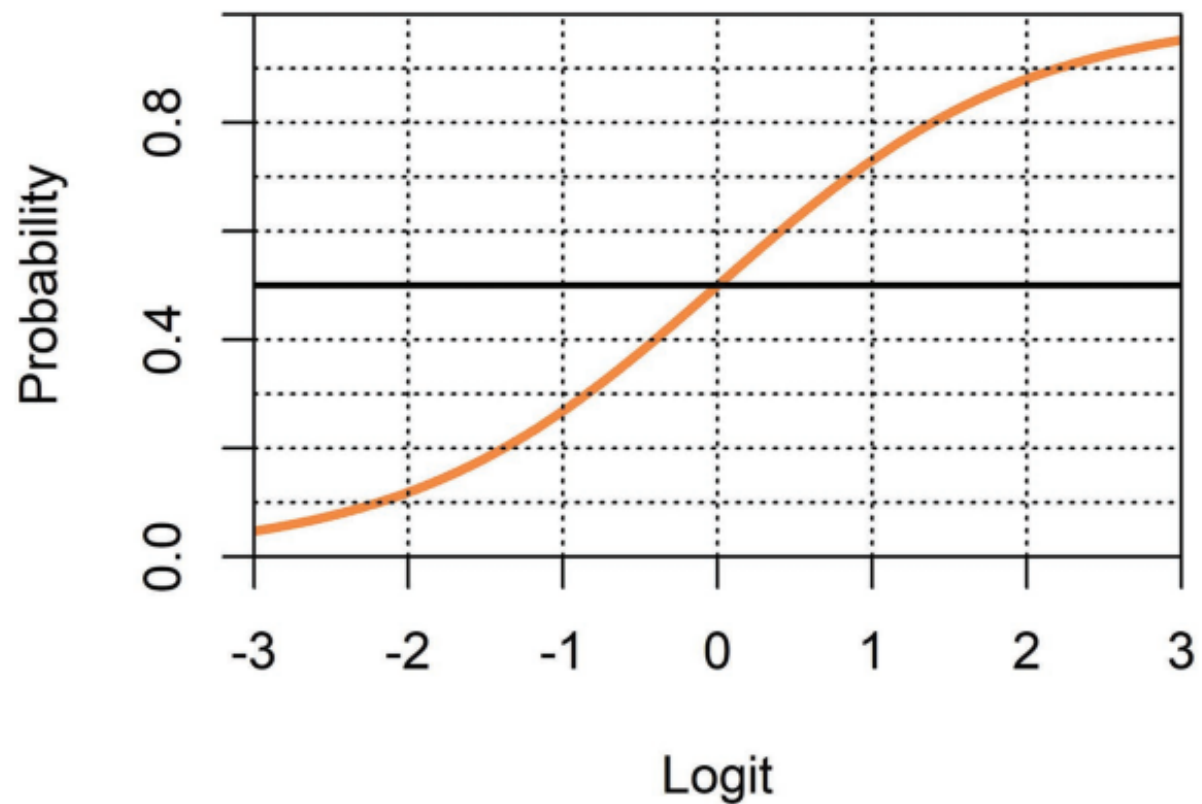
- Logits = El logaritmo de las cuotas,  $\log(\text{cuotas})$ .

$$\text{logit}(p) = \log(p / (1 - p))$$

$$\text{logit}(p) = \log(p) - \log(1 - p)$$

- Útil porque, a diferencia de las probabilidades, los logits pueden tener valores de menos a infinito positivo.

# Logits and Probability



# Logits and Probability

$$P(y=1) = \text{logit}^{-1}(z) = \frac{e^z}{1+e^z} \quad \rightarrow \quad P(\text{success}) = \frac{N_{\text{success}}}{N_{\text{failures}} + N_{\text{success}}} = \frac{N_{\text{success}}}{N_{\text{total}}}$$

$$\frac{e^0}{1+e^0} = \frac{1}{1+1} = 0.5$$

$$\frac{e^{-3}}{1+e^{-3}} = \frac{0.05}{1+0.05} = 0.0474$$

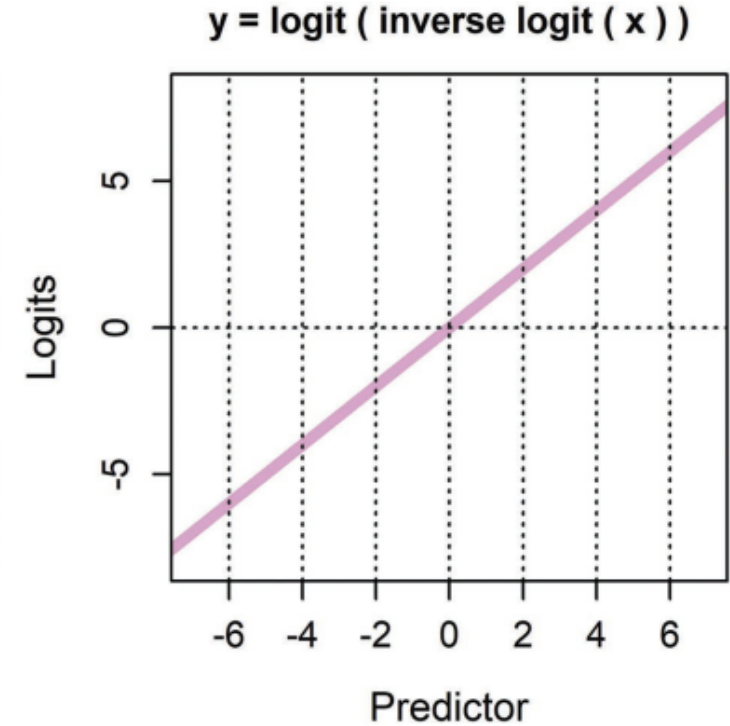
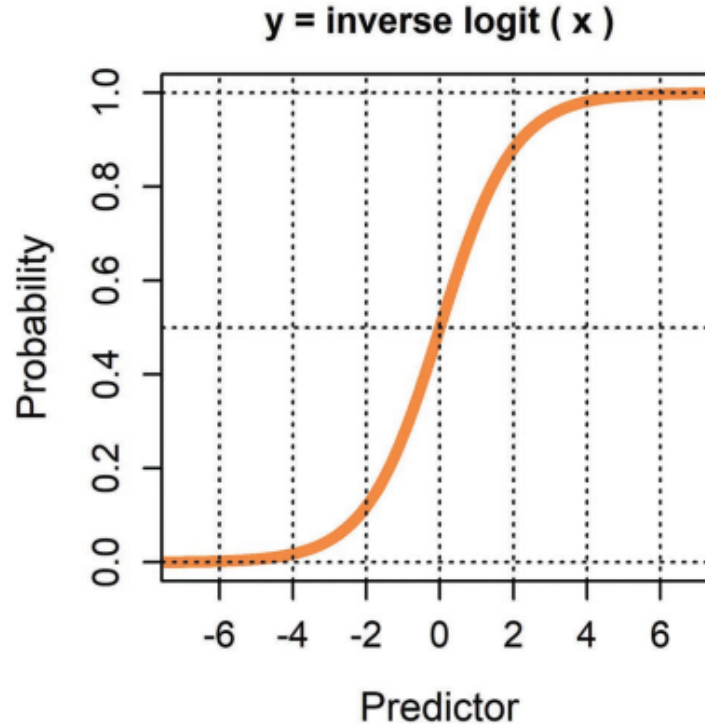
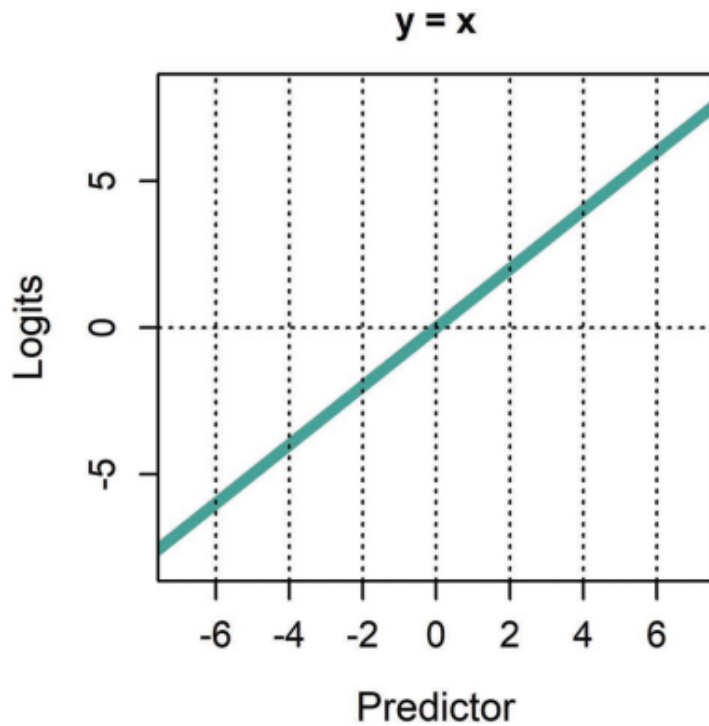
$$\frac{e^3}{1+e^3} = \frac{20.1}{1+20.1} = 0.953$$

# Logits: Odds n' Ends

- A probability of 0.5 is 0 logits. Positive logits mean more likely to be a success, negative logits mean more likely to be a failure.
- $-3$  and  $3$  are 4.7% and 95.2%. Basically  $-3$  and  $3$  logits are useful bounds for “very likely 1” and “very likely 0”.
- Since a logit of 3 translates to a  $P$  of about 0.95, all of the space between  $+3$  and infinity logits represents the probability space between 0.95 and 1, while logits between 0 and 3 represent the space from 0.5 to 0.95.
- Logits far beyond 3 might not have much practical significance. A logit of 4 is a probability of 0.982 and a logit of 6 is 0.997. For many purposes, probabilities of 0.95, 0.98, and 0.99 are nearly interchangeable. Also, it is very difficult to distinguish 95%, 98%, and 99% in practice since you will be observing very few cases of 0.
- Effects can be considered important or not based on how far they get you along  $-3$  to  $3$  (or  $-4$  to  $4$ ). Basically, anything in the  $+1$  range is very likely to matter, while effects smaller than 0.1 or so are likely to have only a small influence on outcomes.

# Prediction in Logits

$$p = \text{logit}^{-1}(\text{Intercept} + \beta \cdot x_{[i]}) = \frac{e^{(\text{Intercept} + \beta \cdot x_{[i]})}}{1 + e^{(\text{Intercept} + \beta \cdot x_{[i]})}}$$
$$F \sim \text{Bernoulli}\left(\frac{e^{(\text{Intercept} + \beta \cdot x_{[i]})}}{1 + e^{(\text{Intercept} + \beta \cdot x_{[i]})}}\right)$$





# Data

```
library (brms)
library (bmmb)
options (contrasts = c('contr.sum', 'contr.sum'))

data (exp_data)

# our dependent variable
exp_data$Female = as.numeric (exp_data$G == 'f')

# make a copy of vtl
exp_data$vtl_original = exp_data$vtl

# center vtl
exp_data$vtl = exp_data$vtl - mean (exp_data$vtl)
```

- **L**: A number from 1 to 15 indicating which *listener* responded to the trial.
- **S**: A number from 1 to 139 indicating which *speaker* produced the trial stimulus.
- **vtl**: An estimate of the speaker's *vocal-tract length* in centimeters.
- **G**: The *apparent gender* of the speaker indicated by the listener, **f** (female) or **m** (male).
- **A**: The *apparent age* of the speaker indicated by the listener, **a** (adult) or **c** (child).



# Research Questions

- (Q1) What is the relationship between speaker VTL and the perception of femaleness?
- (Q2) Does the relationship between VTL and apparent speaker gender vary in an age-dependent manner?

# Description of the Model

$$\text{Female} \sim \text{vtl} * A + (\text{vtl} * A | L) + (1 | S)$$

We're treating our femaleness judgments (1 or 0 for female or male) as coming from a Bernoulli distribution with a probability ( $p$ ) that varies from trial to trial. The logit of the probability ( $z$ ) varies along lines. The lines are specified by intercepts ( $a$ ) and slopes ( $b$ ) that vary from trial to trial, and there is a single continuous predictor (speaker VTL). The intercept of these lines varies based on an overall intercept, an effect for apparent age ( $A$ ), listener-specific effects for apparent age ( $A:L$ ), listener-specific deviations ( $L$ ), and speaker-specific deviations ( $S$ ). The slope of these lines varies based on an overall slope ( $VTL$ , the main effect), deviations based on apparent age ( $VTL:A$ ), listener-specific deviations ( $VTL:L$ ), and listener-specific interactions between apparent age and VTL ( $A:VTL:L$ ). The speaker intercept ( $S$ ) terms were drawn from a normal distribution with a mean of zero and a standard deviation estimated from the data. The listener random effects were drawn from a multivariate normal distribution with means of zero and a covariance matrix estimated from the data. All other effects (e.g., the Intercept, VTL,  $A$ , etc.) were treated as 'fixed' and drawn from prior distributions appropriate for their expected range of values.

$$\text{Female}_{[i]} \sim \text{Bernoulli}(p_{[i]})$$

$$p_{[i]} = \text{logit}^{-1}(z_{[i]})$$

$$z_{[i]} = a_{[i]} + b_{[i]} \cdot \text{vtl}_{[i]}$$

$$a_{[i]} = \text{Intercept} + A + A:L_{[L_{[i]}]} + L_{[L_{[i]}]} + S_{[S_{[i]}]}$$

$$b_{[i]} = VTL + VTL:A + VTL:L_{[L_{[i]}]} + VTL:A:L_{[L_{[i]}]}$$

Priors:

$$S_{[\cdot]} \sim \text{Normal}(0, \sigma_S)$$

$$\begin{bmatrix} L_{[\cdot]} \\ A:L_{[\cdot]} \\ VTL:L_{[\cdot]} \\ VTL:A:L_{[\cdot]} \end{bmatrix} \sim \text{MVNormal} \left( \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma \right)$$

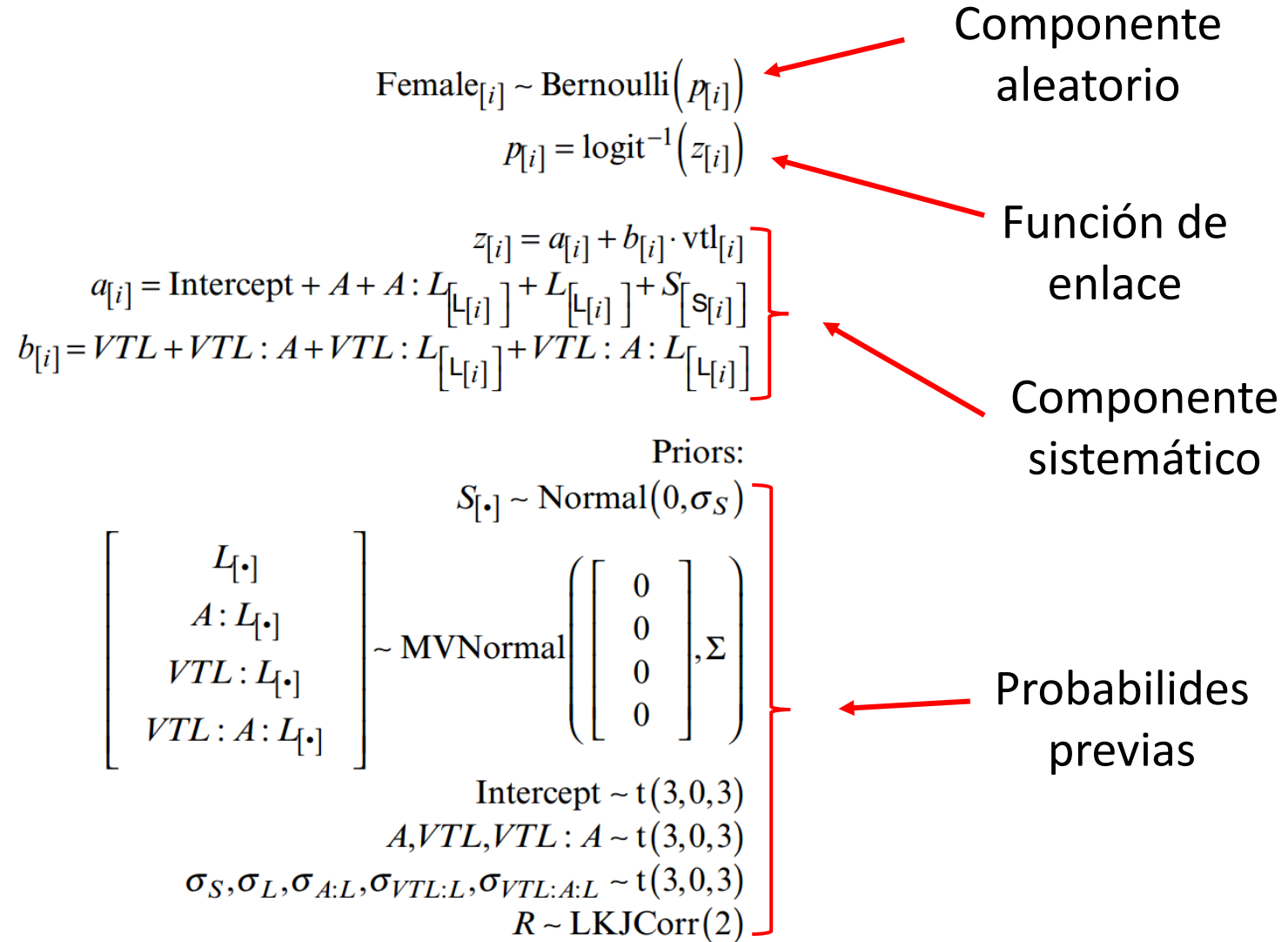
$$\text{Intercept} \sim t(3, 0, 3)$$

$$A, VTL, VTL:A \sim t(3, 0, 3)$$

$$\sigma_S, \sigma_L, \sigma_{A:L}, \sigma_{VTL:L}, \sigma_{VTL:A:L} \sim t(3, 0, 3)$$

$$R \sim \text{LKJCorr}(2)$$

# Description of the Model



# Fitting the Model

```
# Fit the model yourself
priors = c(brms::set_prior("student_t(3, 0, 3)", class = "Intercept"),
           brms::set_prior("student_t(3, 0, 3)", class = "b"),
           brms::set_prior("student_t(3, 0, 3)", class = "sd"),
           brms::set_prior("lkj_corr_cholesky (2)", class = "cor"))

model_gender_vtl =
  brm (Female ~ vtl*A + (vtl*A|L) + (1|S), data=exp_data, chains=4, cores=4,
      family="bernoulli", warmup=1000, iter= 5000, thin = 4,prior=priors)
```



# Interpreting the Model

- ¡Todo está en logits ahora!
- No hay ni sigma ni nu.

```
short_summary(model_gender_vtl)
## Formula: Female ~ vtl * A + (vtl * A | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##
##               Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)      0.55      0.27   0.08   1.14
## sd(vtl)            1.04      0.31   0.53   1.78
## sd(A1)             0.95      0.31   0.44   1.61
## sd(vtl:A1)         0.43      0.24   0.04   0.96
## cor(Intercept,vtl) -0.06      0.32  -0.66   0.57
## cor(Intercept,A1)  -0.22      0.32  -0.78   0.44
## cor(vtl,A1)        -0.46      0.24  -0.82   0.09
## cor(Intercept,vtl:A1) -0.26      0.37  -0.83   0.55
## cor(vtl,vtl:A1)     0.24      0.34  -0.48   0.78
## cor(A1,vtl:A1)     -0.26      0.33  -0.81   0.48
##
## ~S (Number of levels: 139)
##
##               Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)      1.2      0.16   0.91   1.55
##
## Population-Level Effects:
##               Estimate Est.Error 1-95% CI u-95% CI
## Intercept        0.86      0.31   0.27   1.49
## vtl              -3.59      0.39  -4.41  -2.88
## A1               2.65      0.35   2.01   3.39
## vtl:A1           -1.76      0.28  -2.34  -1.24
```



# Additivity

- Las funciones son aditivas si conservan esta relación.

$$\longrightarrow f(x + y) = f(x) + f(y)$$

- La función logit inversa no es aditiva.

$$\longrightarrow \text{logit}^{-1}(\text{Intercept} + A1) \neq \text{logit}^{-1}(\text{Intercept}) + \text{logit}^{-1}(A1)$$

- Significa que tenemos que combinar parámetros antes de transformarlos.

```
# intercept + adult (good)
inverse_logit (0.86 + 2.65)
## [1] 0.971
```

```
# intercept + adult (bad)
inverse_logit (0.86) + inverse_logit (2.65)
## [1] 1.637
```



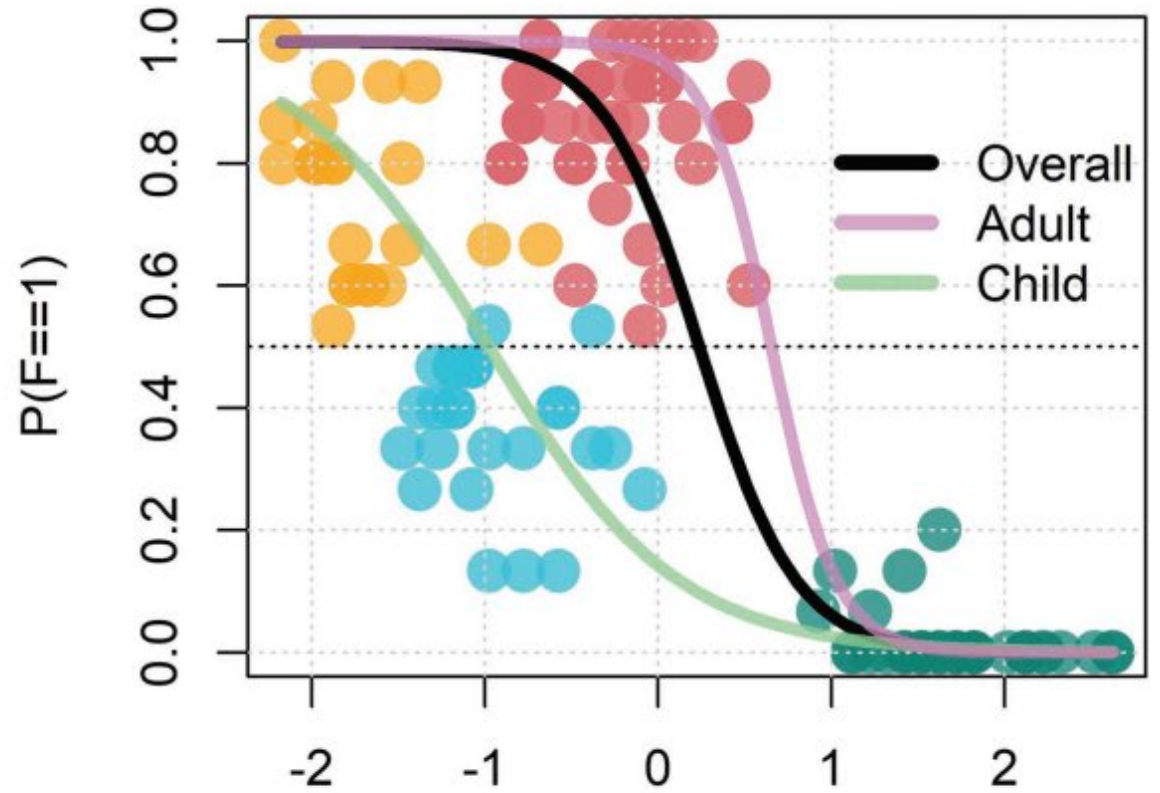
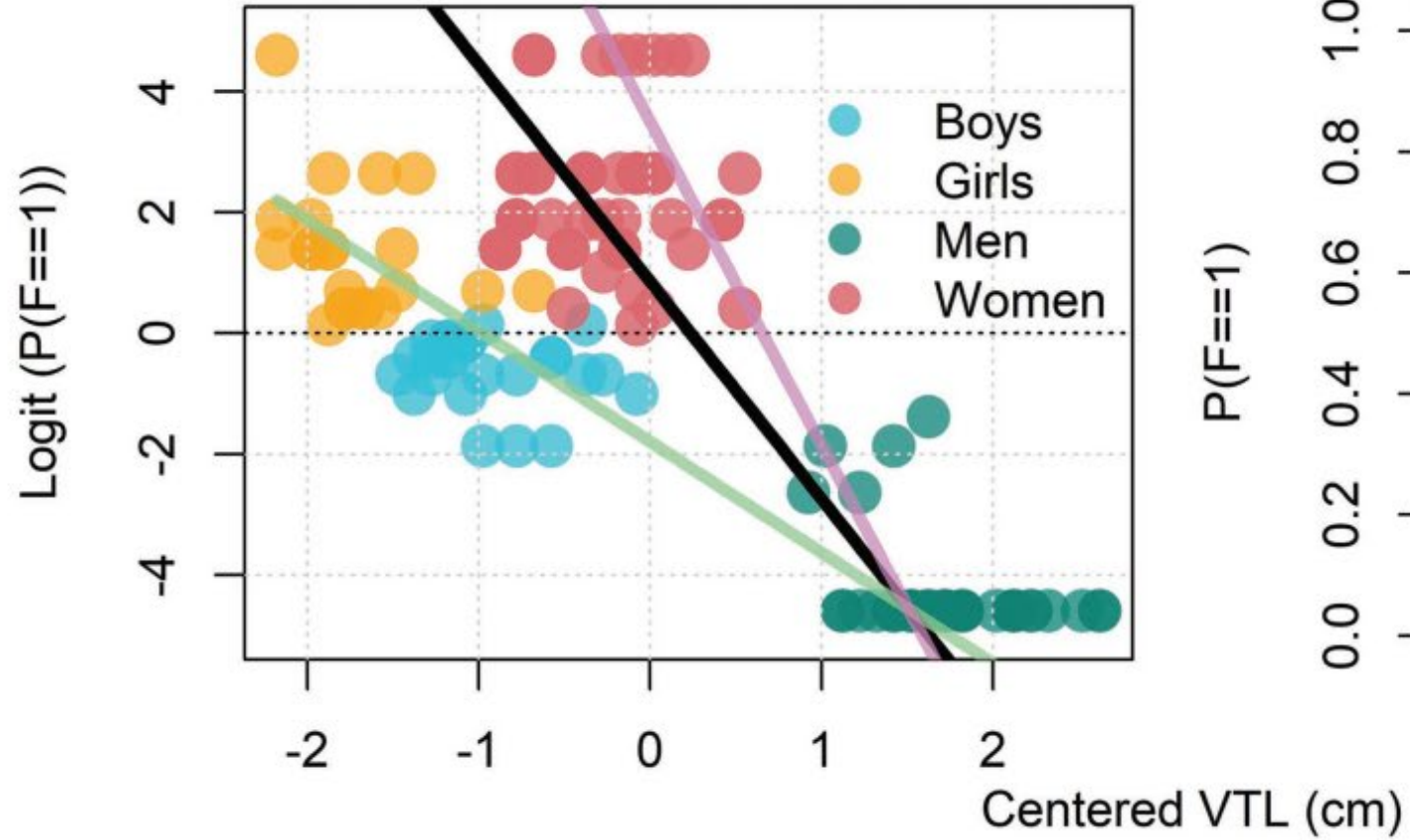
# Simple Effects

```
gender_vtl_hypothesis = bmb::short_hypothesis (
  model_gender_vtl,
  hypothesis = c("Intercept = 0",           # overall intercept
                 "Intercept + A1 = 0",      # adult intercept
                 "Intercept - A1 = 0",      # child intercept
                 "vtl = 0",                 # overall slope
                 "vtl + vtl:A1 = 0",        # adult slope
                 "vtl - vtl:A1 = 0") )      # child slope
```

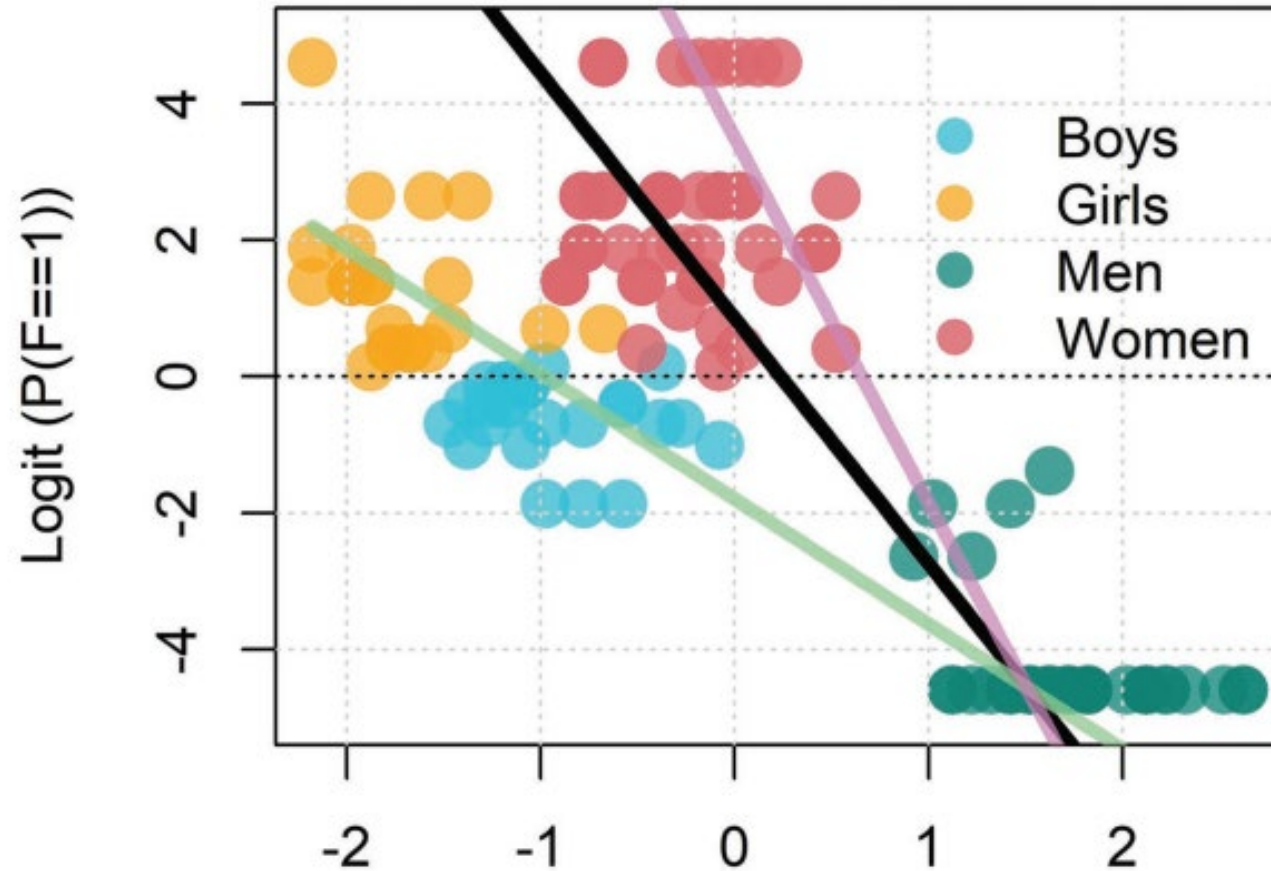
gender\_vtl\_hypothesis

| ## |    | Estimate | Est.Error | Q2.5    | Q97.5   | hypothesis         |
|----|----|----------|-----------|---------|---------|--------------------|
| ## | H1 | 0.8563   | 0.3084    | 0.2672  | 1.4934  | (Intercept) = 0    |
| ## | H2 | 3.5035   | 0.5022    | 2.5716  | 4.5727  | (Intercept+A1) = 0 |
| ## | H3 | -1.7909  | 0.4337    | -2.6816 | -0.9702 | (Intercept-A1) = 0 |
| ## | H4 | -3.5911  | 0.3868    | -4.4145 | -2.8822 | (vtl) = 0          |
| ## | H5 | -5.3493  | 0.5664    | -6.5654 | -4.3457 | (vtl+vtl:A1) = 0   |
| ## | H6 | -1.8328  | 0.3681    | -2.5856 | -1.1353 | (vtl-vtl:A1) = 0   |

# Simple Effects



# Classification: the y intercept



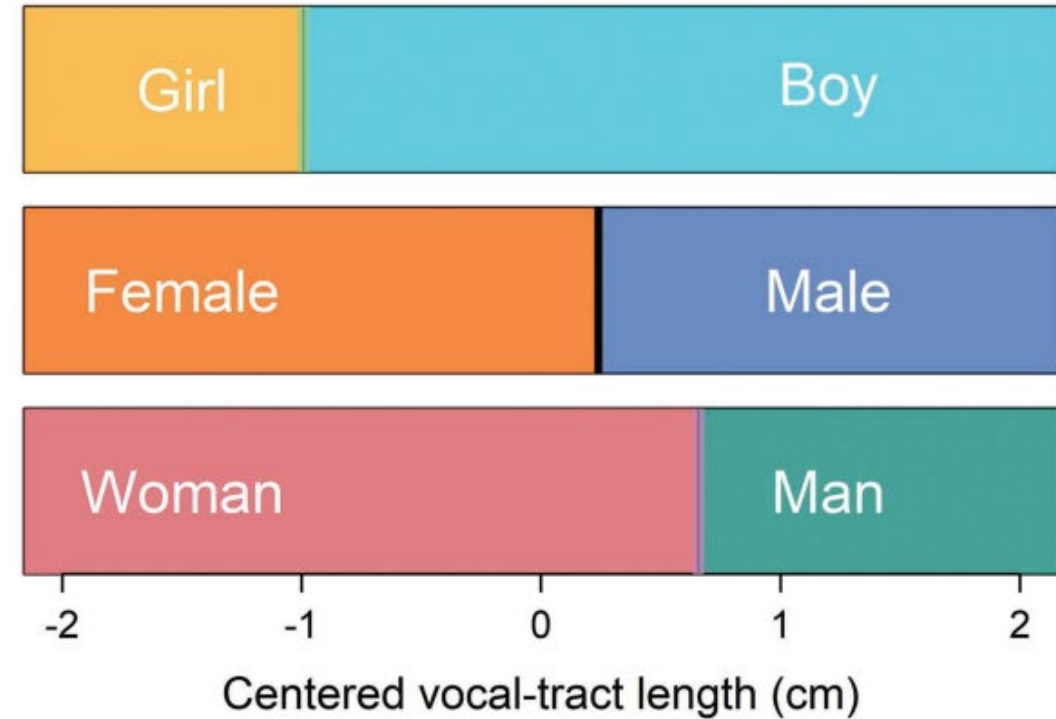
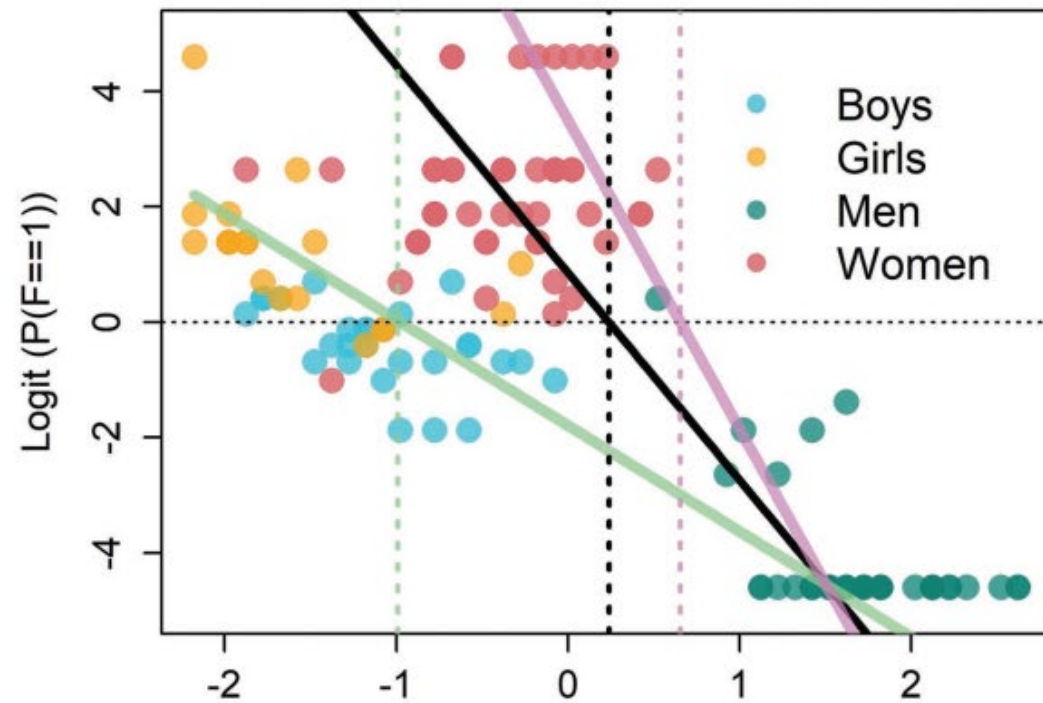
$$y = a + b \cdot x$$

$$0 = a + b \cdot x$$

$$-a = b \cdot x$$

$$-a / b = x$$

# Classification: the y intercept





# Classification: the y intercept

```
samples = fixef (model_gender_vt1, summary = FALSE)
```

```
# calculate overall boundary = -a/b  
boundary = -samples[, "Intercept"] / samples[, "vt1"]
```

```
# same but for adults  
boundary_adults = -(samples[, "Intercept"] + samples[, "A1"]) /  
  (samples[, "vt1"] + samples[, "vt1:A1"])  
  
# now for children  
boundary_children = -(samples[, "Intercept"] - samples[, "A1"]) /  
  (samples[, "vt1"] - samples[, "vt1:A1"])
```

$$y = a + b \cdot x$$

$$0 = a + b \cdot x$$

$$-a = b \cdot x$$

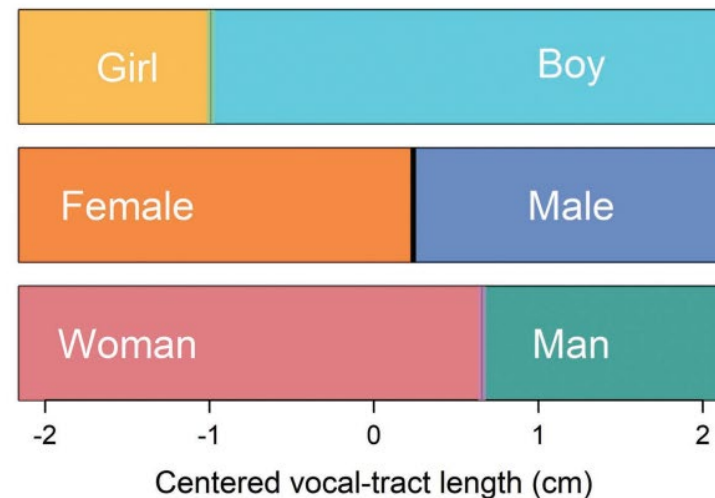
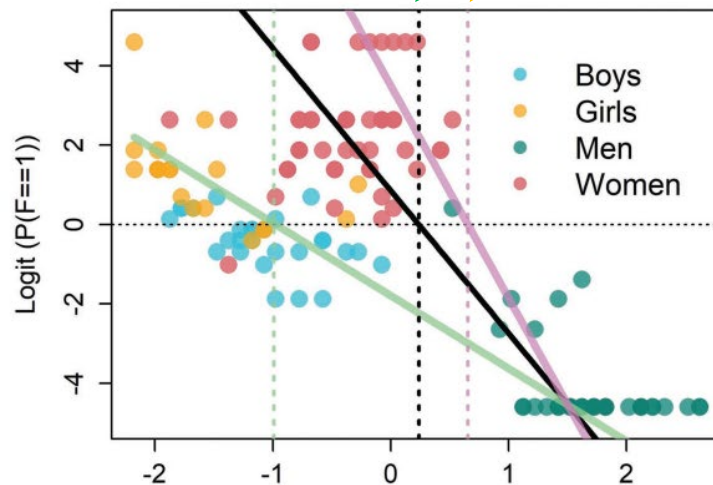
$$-a / b = x$$

# Boundaries

```
boundaries_1 = posterior_summary (  
  cbind (boundary, boundary_adults, boundary_children))
```

```
boundaries_1
```

```
##              Estimate Est.Error      Q2.5      Q97.5  
## boundary      0.2397   0.08613   0.07438   0.4104  
## boundary_adults 0.6558   0.07169   0.51829   0.7958  
## boundary_children -0.9925  0.23018  -1.48021  -0.5908
```





# Answering our Research Questions

- (Q1) What is the relationship between speaker VTL and the perception of femaleness?
- (Q2) Does the relationship between VTL and apparent speaker gender vary in an age-dependent manner?

Speaker VTL is negatively related to the perception of femaleness with a slope of  $-3.59$  logits per unit change in cm (s.d. =  $0.39$ , 95% C.I. =  $[-4.41, -2.88]$ ). This effect increased by about 50% when listeners thought the speaker was an adult and decreased by about 50% when listeners thought the speaker was a child (mean =  $2.65$ , s.d. =  $0.35$ , 95% C.I. =  $[2.01, 3.39]$ ). Our results do indicate that the relationship between VTL and apparent femaleness varies as a function of the apparent age of the speaker. These differences can be understood in terms of the information presented in Figure 10.5 and, in particular, the territorial maps presented in the right plot. In the plot, we see that when listeners thought the speaker was a child, the boundary between male and female speakers is at a lower value of VTL (12.4 cm based on our calculations above), which makes sense given that children are smaller overall. When the listener thinks the speaker is an adult, a higher value of VTL (14.0 cm) is required to predict a male response.

# Signal Detection Theory

- Imagina que quieres saber si alguien puede identificar a los hablantes masculinos y femeninos.
- Identifican al 100% de las hablantes femeninas como mujeres.....
- Pero también el 100% de los hombres como mujeres.
- La teoría de detección de señales proporciona un marco para modelar la capacidad de saber cuándo una señal está presente y cuándo no está presente.

# Hits and False Alarms

- **Aciertos** son pruebas cuando el oyente dice que la señal está presente, y así es.
- **Falsas alarmas** son pruebas cuando el oyente dice que la señal está presente, y no lo está.

$$H = P(\text{Female} = 1 \mid G_v = f)$$

$$\text{FA} = P(\text{Female} = 1 \mid G_v = m)$$

```
tapply(exp_data$Female, exp_data$G_v, mean)
##           f           m
## 0.8219 0.1694
```

# Sensitivity

- La sensibilidad es la capacidad de discriminar categorías.
- La diferencia entre aciertos y falsas alarmas: más aciertos + menos falsas alarmas = mayor discriminación.
- Usar probabilidades es malo. En su lugar, usamos la diferencia en logits.

$$d = \text{logit}(H) - \text{logit}(FA)$$

“d”



$$d' = z(H) - z(FA)$$

“d-prime”



# Bias

- El sesgo de respuesta es la tendencia a seleccionar una categoría más que otra.
- La suma de aciertos y falsas alarmas: más aciertos + más falsas alarmas = mayor sesgo.

$$b = -c' = \frac{1}{2} [\text{logit}(H) + \text{logit}(FA)]$$

  
“sesgo”

$$c' = -\frac{1}{2} [\text{logit}(H) + \text{logit}(FA)]$$

  
“criterio”



# Sensitivity and Bias

- Ejemplo 1: Imagina que una persona puede adivinar cuándo un lanzamiento de moneda será cara el 90% de las veces. ¿No deberían ser también capaces de identificar las colas el 90% de las veces?
- Ejemplo 2: Imagine que una persona identifica al 80% de las mujeres como mujeres y al 60% de los hombres como hombres. Si saben que las mujeres son mujeres el 80% del tiempo, también saben que las mujeres no son hombres el 80% del tiempo. ¿Cómo es posible que solo tengan un 60% de razón para los hombres?
- La "precisión" desigual para dos categorías no refleja una sensibilidad diferente. Refleja un sesgo de respuesta.

# Example: Gender Detection in Children

```
# adult speaker data
adults = exp_data[exp_data$A_v == "a",]

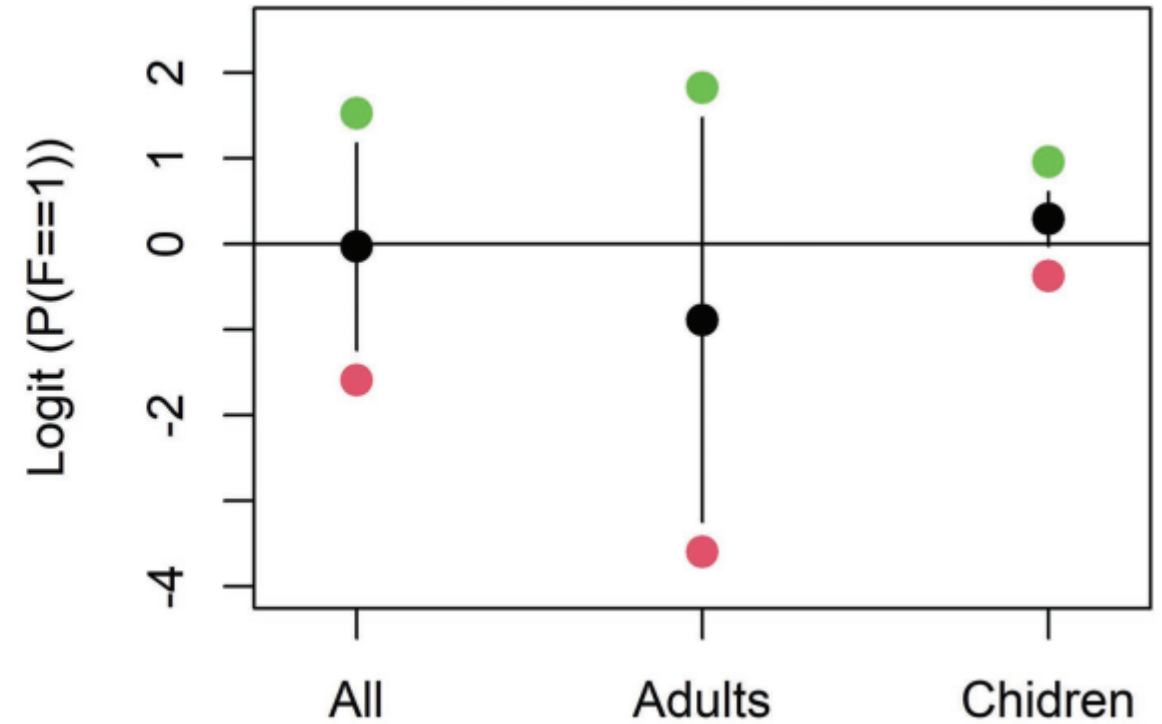
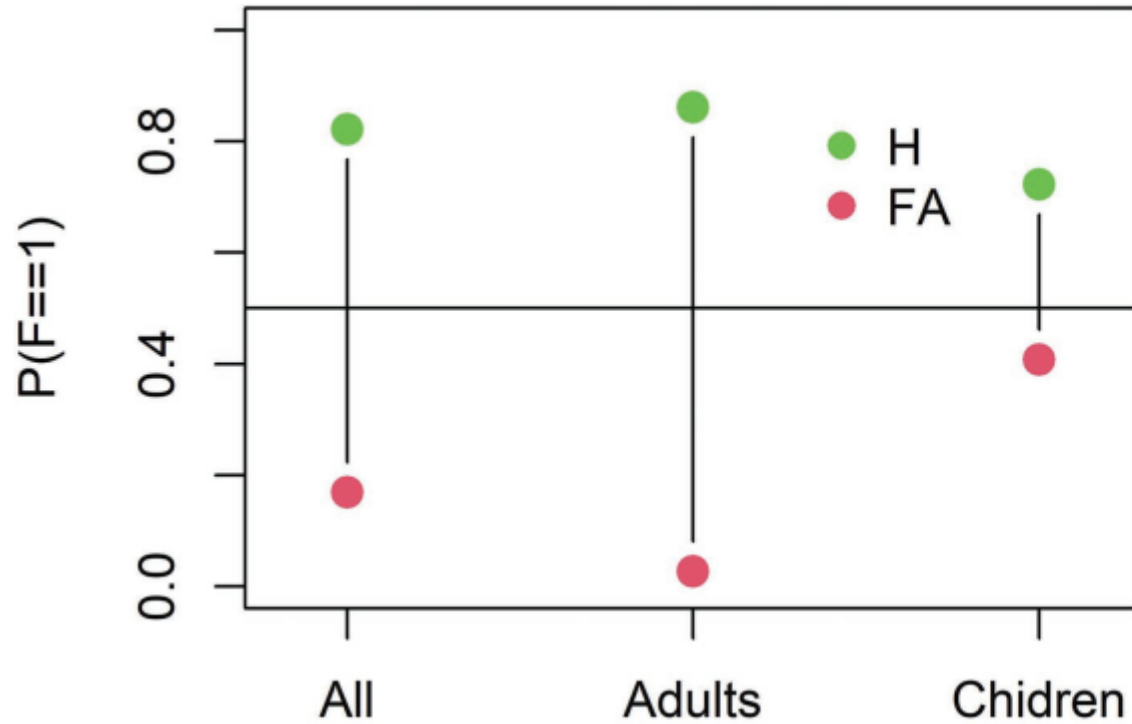
# child speaker data
children = exp_data[exp_data$A_v == "c",]
```

```
# hit and false alarm rate, overall
tapply (exp_data$Female, exp_data$G_v, mean)
##           f           m
## 0.8219 0.1694

# hit and false alarm rate, for adult
tapply (adults$Female, adults$G_v, mean)
##           f           m
## 0.86111 0.02667

# hit and false alarm rate, for children
tapply (children$Female, children$G_v, mean)
##           f           m
## 0.7228 0.4074
```

# Example: Gender Detection in Children



# Data and Research Questions

```
library (brms)
library (bmmmb)
options (contrasts = c('contr.sum', 'contr.sum'))
data (exp_data)

# our dependent variable
exp_data$Female = as.numeric (exp_data$G == 'f')

# make a copy of vtl
exp_data$vtl_original = exp_data$vtl

# center vtl
exp_data$vtl = exp_data$vtl - mean (exp_data$vtl)

# create veridical gender predictor
exp_data$F_v = ifelse (exp_data$G_v=="f", 1, -1)
```

# Data and Research Questions

- **L**: A number from 1 to 15 indicating which *listener* responded to the trial.
- **S**: A number from 1 to 139 indicating which *speaker* produced the trial stimulus.
- **G**: The *apparent gender* of the speaker indicated by the listener, **f** (female) or **m** (male).
- **G\_v**: The *veridical gender* of the speaker indicated by the listener, **f** (female) or **m** (male).
- **A\_v**: The *veridical age* of the speaker indicated by the listener, **a** (adult) or **c** (child).

(Q1) How different is listeners' ability to discriminate the gender of children and adults?

(Q2) Is response bias different for children and for adults?



# Description of the Model

Para crear un modelo de teoría de detección de señales, debe incluir la categoría verídica como predictor.


$$\text{Female} \sim F\_v * A\_v + (F\_v * A\_v | L) + (1 | S)$$

Los términos que interactúan con la categoría verídica reflejan sensibilidad.


$$\text{Female} \sim F\_v + A\_v + F\_v:A\_v + (F\_v + A\_v + F\_v:A\_v | L) + (1 | S)$$


Los términos que no interactúan con la categoría verídica reflejan sesgo.

# Description of the Model

$$\text{Female}_{[i]} \sim \text{Bernoulli}(p_{[i]})$$

$$p_{[i]} = \text{logistic}(z_{[i]})$$

Términos de  
sesgo

Términos de  
sensibilidad

$$z_{[i]} = b_{[i]} + d_{[i]}$$

$$b_{[i]} = \text{Intercept} + A_v + A_v : L_{[L[i]]} + L_{[L[i]]} + S[S[i]]$$

$$d_{[i]} = F_v + F_v : A_v + F_v : L_{[L[i]]} + F_v : A_v : L_{[L[i]]}$$

Priors:

$$S_{[\cdot]} \sim \text{Normal}(0, \sigma_S)$$

$$\begin{bmatrix} L_{[\cdot]} \\ A_v : L_{[\cdot]} \\ F_v : L_{[\cdot]} \\ A : F_v : L_{[\cdot]} \end{bmatrix} \sim \text{MVNormal} \left( \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma \right)$$

$$\text{Intercept} \sim t(3, 0, 3)$$

$$A, VTL, A : VTL \sim t(3, 0, 3)$$

$$\sigma_L, \sigma_{A_v:L}, \sigma_{F_v:L}, \sigma_{A_v:F_v:L}, \sigma_S \sim t(3, 0, 3)$$

$$R \sim \text{LKJCorr}(2)$$

# Fitting the Model

```
# Fit the model yourself
model_gender_dt =
  brm (Female ~ F_v*A_v + (F_v*A_v|L) + (1|S), data=exp_data,
      chains=4, cores=4, family="bernoulli",
      warmup=1000, iter = 5000, thin = 4,
      prior = c(set_prior("student_t(3, 0, 3)", class = "Intercept"),
                set_prior("student_t(3, 0, 3)", class = "b"),
                set_prior("student_t(3, 0, 3)", class = "sd"),
                set_prior("lkj_corr_cholesky (2)", class = "cor")))
```

# Interpreting the Model: Fixed Effects

```
fixef (model_gender_dt)
##           Estimate Est.Error      Q2.5      Q97.5
## Intercept   -0.3514    0.2634  -0.8774    0.1545
## F_v         2.2522    0.1851   1.9038    2.6424
## A_v1        -0.7712    0.2255  -1.2340   -0.3309
## F_v:A_v1     1.3446    0.1764   1.0086    1.7019
```

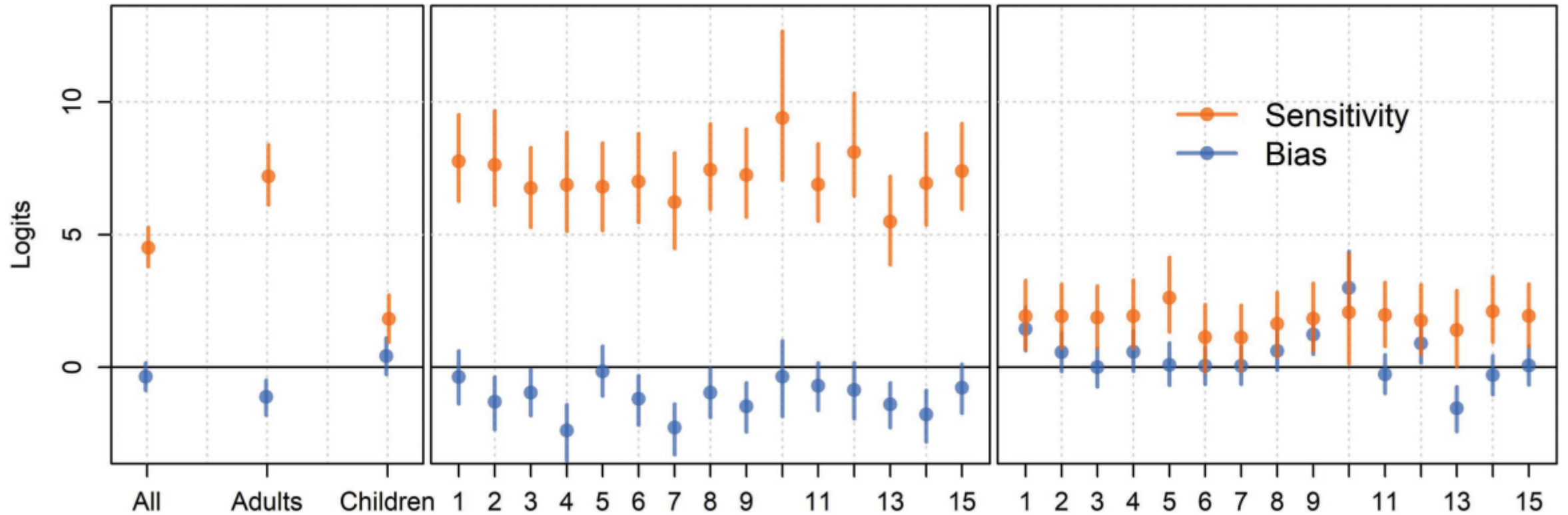
# Simple Effects: Sensitivity and Bias

```
gender_dt_hypothesis = bmb::short_hypothesis (  
  model_gender_dt,  
  hypothesis = c("Intercept = 0",           # overall bias  
                 "Intercept + A_v1 = 0",     # adult bias  
                 "Intercept - A_v1 = 0",     # child bias  
                 "2*(F_v) = 0",              # overall sensitivity  
                 "2*(F_v + F_v:A_v1) = 0",   # adult sensitivity  
                 "2*(F_v - F_v:A_v1) = 0"))  # child sensitivity
```

```
# listener-dependent biases for veridical adults  
biases_adult = bmb::short_hypothesis (  
  model_gender_dt,  
  hypothesis = c("Intercept+A_v1 = 0"), group="L", scope="coef")
```



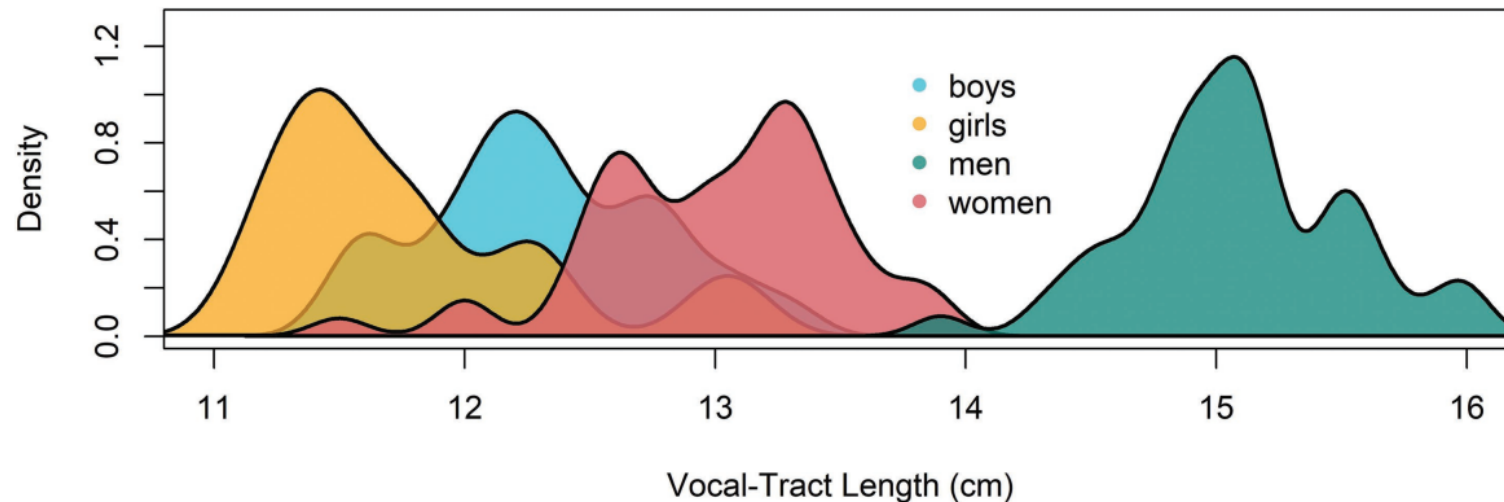
# Simple Effects: Sensitivity and Bias



# Answering our Research Questions

Table 10.1 Posterior means, standard errors, and 2.5% and 97.5% quantiles for bias and sensitivity under different conditions

|                     | <i>Estimate</i> | <i>Est.error</i> | <i>Q2.5</i> | <i>Q97.5</i> |
|---------------------|-----------------|------------------|-------------|--------------|
| Overall bias        | −0.35           | 0.26             | −0.88       | 0.15         |
| Adult bias          | −1.12           | 0.34             | −1.81       | −0.49        |
| Child bias          | 0.42            | 0.36             | −0.28       | 1.11         |
| Overall sensitivity | 4.50            | 0.37             | 3.81        | 5.28         |
| Adult sensitivity   | 7.19            | 0.57             | 6.13        | 8.39         |
| Child sensitivity   | 1.82            | 0.44             | 0.94        | 2.72         |



```
xtabs ( ~ adults$C_v + adults$C)
##               adults$C
## adults$C_v    b     g     m     w
##             b     0     0     0     0
##             g     0     0     0     0
##             m    31     0   626    18
##             w    97   109     3   511
```